

# Xingyue Huang

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[Google Scholar](#)

## Education

University of Oxford

2023 – 2026

DPhil in Computer Science

Advisors: Prof. Michael Bronstein (DeepMind Chair of AI), Prof. Ismail Ilkan Ceylan

University of Oxford

2019 – 2023

MMathCompSci in Mathematics and Computer Science

Graduated with Distinction

## Professional Experience

Research Scientist Intern

06/2026 – 09/2026

Meta Platforms Inc.

- Developed an LLM-based survey imputation system for advertiser campaign measurement, modeling responses from 5k advertisers across 18 questions and 10 weekly waves.
- Reduced aggregate distributional error by  $3.4\times$ , from 26.3% to 7.8%, using quantile-mapped KNN on Llama 4.
- Designed LLM-XGBoost ensemble methods achieving the best distributional fidelity, reducing PPE to 6.7% and improving accuracy by 1.7% by using LLM predictions as tabular features.

Predoctoral Fellow

03/2026 – 05/2026

AITHYRA Research Institute for Biomedical Artificial Intelligence

- Developed *RelAgent*, an LLM-based autonomous data scientist that searches over SQL feature programs and model choices for relational learning.
- Achieved rank 1.00 on RelBenchV2 and 4DBInfer classification benchmarks, improving average AUROC over KumoRFM-v2 from 85.91 to 87.32 and from 79.96 to 81.38, respectively.
- Designed deterministic inference without further LLM calls, yielding interpretable SQL-defined feature maps paired with classical predictors for scalable database deployment.

Research Intern

06/2025 – 10/2025

Snap Inc.

- Developed Threshold Differential Attention (TDA), a sink-free, ultra-sparse attention mechanism for long-context LLMs, achieving  $>99\%$  exact zeros while matching Softmax accuracy on QA benchmarks.
- Pre-trained large language models from scratch and demonstrated long-context robustness on SCROLLS passkey retrieval, where TDA outperformed Softmax by approximately  $2.5\times$  at 4k-token contexts.
- Co-authored *Hierarchical Token Prepending*, improving long-document embeddings in decoder-only LLMs via block-level summary tokens, with 5% gains across 11 retrieval datasets and 30 embedding benchmarks.

Research Intern

10/2024 – 06/2025

Eigent-AI

- Built a tool-use synthetic data generation pipeline producing 20k verified execution traces for CAMEL-AI.
- Used back-translated tool trajectories for supervised fine-tuning, improving math benchmark accuracy by 5%.
- Led the *Loong* verifier-driven RL framework for long chain-of-thought synthesis.

## Publications

### *LLM, Agents, and Pre/Post-Training:*

[RelAgent: LLM Agents as Data Scientists for Relational Learning](#), Xingyue Huang, Louis Tichelman, Jinwoo Kim, Krzysztof Olejniczak, and İsmail İlkan Ceylan, *NeurIPS*, 2026.

[Threshold Differential Attention for Sink-Free, Ultra-Sparse, and Non-Dispersive Language Modeling](#), Xingyue Huang\*, Xueying Ding\*, Mingxuan Ju, Yozen Liu, Neil Shah, and Tong Zhao, *ACL*, 2026 (Oral).

[Hierarchical Token Prepending: Enhancing Information Flow in Decoder-based LLM Embeddings](#), Xueying Ding\*, Xingyue Huang\*, Mingxuan Ju, Liam Collins, Yozen Liu, Leman Akoglu, Neil Shah, and Tong Zhao, *ACL*, 2026 (Oral).

[Loong: Synthesize Long Chain-of-Thoughts at Scale through Verifiers](#), Xingyue Huang\*, Rishabh\*, Gregor Franke\*, Ziyi Yang\*, Jiamu Bai, Weijie Bai, Jinhe Bi, Zifeng Ding, Yiqun Duan, Chengyu Fan, *et al.*, *LAW Workshop at NeurIPS*, 2025.

[Distilling Tool Knowledge into Language Models via Back-Translated Traces](#), Xingyue Huang\*, Xianglong Hu\*, Zifeng Ding\*, Yuan He, Rishabh, Waleed Alzarooni, Ziyu Ye, Wendong Fan, Bailan He, Haige Bo, Changran Hu, and Guohao Li, *Multi-Agent Systems in the Era of Foundation Models Workshop at ICML*, 2025.

[Self-Exploring Language Models for Explainable Link Forecasting on Temporal Graphs via Reinforcement Learning](#), Zifeng Ding, Shenyang Huang, Zeyu Cao, Emma Kondrup, Zachary Yang, **Xingyue Huang**, Yuan Sui, Zhangdie Yuan, Yuqicheng Zhu, Xianglong Hu, Yuan He, Farimah Poursafaei, Michael Bronstein, and Andreas Vlachos, *Workshop on New Perspectives in Graph Machine Learning, NeurIPS 2025*.

[What are the Right Symmetries for Formal Theorem Proving?](#), Krzysztof Olejniczak, Radoslav Dimitrov, **Xingyue Huang**, Bernardo Cuenca Grau, Jinwoo Kim, and İsmail İlkan Ceylan, *AI for Math Workshop at ICML, 2026*.

### **Graph and Relational Learning:**

[HYPER: A Foundation Model for Inductive Link Prediction with Knowledge Hypergraphs](#), **Xingyue Huang**, Mikhail Galkin, Michael M. Bronstein, and İsmail İlkan Ceylan, *ICLR, 2026*.

[Flock: A Knowledge Graph Foundation Model via Learning on Random Walks](#), Jinwoo Kim\*, **Xingyue Huang\***, Krzysztof Olejniczak, Kyungbin Min, Michael Bronstein, Seunghoon Hong, and İsmail İlkan Ceylan, *ICLR, 2026*.

[How Expressive are Knowledge Graph Foundation Models?](#), **Xingyue Huang**, Pablo Barceló, Michael M. Bronstein, İsmail İlkan Ceylan, Mikhail Galkin, Juan L Reutter, and Miguel Romero Orth, *ICML, 2025*.

[Of Graphs and Tables: Zero-Shot Node Classification with Tabular Foundation Models](#), Adam Hayler, **Xingyue Huang**, İsmail İlkan Ceylan, Michael M. Bronstein, and Ben Finkelshtein, *Workshop on New Perspectives in Graph Machine Learning, NeurIPS 2025 (Oral)*.

[One Model, Any Conjunctive Query: Graph Neural Networks for Answering Queries over Incomplete Knowledge Graphs](#), Krzysztof Olejniczak, **Xingyue Huang**, Mikhail Galkin, and İsmail İlkan Ceylan, *Learning on Graphs Conference, 2025*.

[Cooperative Graph Neural Networks](#), Ben Finkelshtein, **Xingyue Huang**, Michael Bronstein, and İsmail İlkan Ceylan, *ICML, 2024*.

[Theoretical Insights into Line Graph Transformation on Graph Learning](#), Fan Yang and **Xingyue Huang**, *Workshop on Symmetry and Geometry in Neural Representations at NeurIPS, 2024*.

[A Theory of Link Prediction via Relational Weisfeiler-Leman on Knowledge Graphs](#), **Xingyue Huang**, Miguel Romero Orth, İsmail İlkan Ceylan, and Pablo Barceló, *NeurIPS, 2023*.

[Link Prediction with Relational Hypergraphs](#), **Xingyue Huang**, Miguel Romero Orth, Pablo Barceló, Michael M. Bronstein, and İsmail İlkan Ceylan, *Transactions on Machine Learning Research, 2025*.

### **Awards**

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- Oriel Student Scholarships for Academic Merit, 2021 – 2023.
- NeurIPS 2025 Top Reviewer Award.
- ICML 2026 Silver Reviewer Award.

### **Service**

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**Reviewer:** NeurIPS 2025, 2026; ICML 2026; ICLR 2026, 2027; TMLR

#### **Invited Talks:**

- *RelAgent: LLM Agents as Data Scientists for Relational Learning*; Stanford University; CAMEL-AI.
- *Graph Foundation Models Tutorial*; Learning on Graphs 2025.
- *How Expressive are Knowledge Graph Foundation Models?*; Snap Inc., 2025.
- *Relational Hypergraphs for Knowledge Graph Foundation Models*; TU Wien, 2026.

**Lead Organizer:** Workshop on Graph Foundation Models: A New Era of Graph Machine Learning, ICML 2026.

**Organizer:** Scaling Environment of Agents (SEA) Workshop, NeurIPS 2025.